What Single-Image Reflection Removal Does to the Occluder Region: A Diagnostic Benchmark on Tinted Glass

Anonymous CVPR submission

Abstract

Single-image reflection removal (SIRR) implicitly assumes a friendly photometric regime: the reflected scene is bright, the transmitted scene is well exposed, and the task is to separate *two comparable layers.*

We show that ***tinted window film destroys this premise****. The film is dark: it attenuates the reflected layer* and *the transmitted layer at the same time. On a hand-annotated* 15*-scene panel shot through automotive window film we find that the film treats the two layers* differently*: the transmitted layer passes through the film bulk and is scattered and dyed, while the reflection comes off the film surface. The result is a layer asymmetry—the transmission is* $2.3\times$ *softer than on ordinary glass and the reflection is* $2.0\times$ *sharper relative to it—which is* not *explained by brightness: mean levels, reflection magnitude and spectral overlap all match Real20. Under such a regime SIRR is not a separation problem but an* identification *problem.*

This reframing yields four measurable failure modes. (i) The reflection is not *strongest inside the occluder region* Ω *but in the annulus around it:* $4.3\times$ *stronger than inside* Ω*, covering* 44% *of the frame, and that annulus is where every model is worst (*19.4–$19.8\,\mathrm{dB}$ *vs.* 23.5 *far away). Inside* Ω *almost nothing needs to change yet every method is worse than doing nothing (identity* $24.52\,\mathrm{dB}$ *vs.* 20.9–23.6 *for six public methods and* 22.4 *for ours), so the residual is read as a* boundary*. (ii) The damage is not a photometric offset: after a per-image affine correction the models retain only* 0.48–0.65 *of the ground-truth structure inside* Ω*. (iii) The reflection and the scene share the same frequency band, so simply restoring the input's high frequencies* lowers *PSNR—faithful recovery requires semantic priors, not better filtering. (iv) What survives fine-tuning is per-image: a per-image affine oracle recovers* 0.15–$0.26\,\mathrm{dB}$ *whereas a single shared affine fitted on other scenes recovers* $0.07\,\mathrm{dB}$*.*

The standard ground-truth protocol—covering the glass with dark cloth—adds a systematic illumination change, so photometric alignment and reflection removal cannot be treated separately.

We give a region-relative loss that is load-bearing ($-4.2\,\mathrm{dB}$ *if removed), a mask-free change-aware regularizer derived from a random-disk control (*$+0.50\,\mathrm{dB}$*, per-scene* 95% *CI* $[+0.34, +0.67]$*, positive on* $14/15$ *scenes, zero inference cost), and a systematic set of negative results including perceptual losses, high-frequency splicing and data mixing.*

1. Introduction

Single-image reflection removal (SIRR) addresses a situation everyone with a camera has met: a photograph taken through a pane of glass, in which the scene behind the camera is superimposed on the scene in front of it. The setting is ubiquitous—car windows, shop and office facades, museum cases, aquaria—and the degradation is severe enough that SIRR has become a standing task in computational photography. The dominant recipes are now data-driven: large paired corpora of "with reflection / without reflection" images, a network that regresses the transmitted layer, and an objective that penalises the difference from the clean reference. Classical layer priors gave way to cascaded and recurrent designs, then to reversible and dual-stream architectures, and each generation has advanced the state of the art on benchmarks such as Real20 and SIR^2.

What these recipes share is an implicit assumption about the *glass*. In deployment the glass is rarely bare: automotive and architectural glazing is almost always *filmed*, for heat rejection, privacy and UV protection. A dark film changes the imaging in a way that the standard assumption does not cover. The film sits on the scene side, so the light we want must travel through it—and is scattered, absorbed and tinted on the way—while the light we want to discard comes off the bare surface facing the camera and passes through nothing. The observed image is therefore not a bright reflection layered on a well-exposed scene, but a *sharp* reflection over a *soft*, desaturated transmission. Methods trained on ordinary glass, and evaluated on ordinary glass, have no reason to expect this, and empirically they do not survive it (Fig. 1).

The community has of course attacked the difficulty of SIRR from several directions: stronger backbones and attention, reversible or invertible connections to preserve information, explicit estimates of the reflection rate, physics-inspired layer decompositions, and larger or more realis-



Figure 1. **Same models, two kinds of glass.** Top, on a standard benchmark (Real20): the state-of-the-art method [10] removes the reflection almost completely, while a second recent method (LightRR [2]) removes only part of it. Bottom, on a pair from our filmed-glass panel: the *same* RDNet does not remove the reflection at all—instead it darkens the frame and strips transmitted detail—and LightRR changes almost nothing. The difference is not the network but the imaging: window film is laminated on the scene side, so the transmitted light passes through the film while the reflection of the behind-camera environment does not. The two layers end up separated in *band* rather than in *level*, and a model trained on ordinary glass misreads what to take away.

tic synthetic corpora. What these efforts have in common is where they are measured: on benchmarks whose reflections are *dense*, covering most of the frame, so that an average error over the image is already a faithful description of the failure. Three things are consequently missing. There is no real paired dataset shot through filmed glass, so the regime cannot even be measured. There is no analysis of *where* a method fails—strong reflection is not distributed uniformly over the frame, and a single average number cannot distinguish the regions where removal is necessary from those where it is harmful. And because no such analysis exists, the losses used in training are inherited from dense-reflection settings, where they work.

This paper takes the opposite order. We first ask what filmed glass actually does to the two layers, and answer it with measurements: the transmission loses high-frequency content relative to ordinary glass while the reflection does not, so the layer that carries the detail and the layer that must be discarded are separated in frequency rather than in brightness. We then ask *where* current methods fail, and find that the answer is unintuitive: the reflection is weakest exactly in the region the eye fixates on—the patch where a dark object of the reflected environment projects, which we call the low-change region—and strongest in the annulus immediately around it. The annulus is large, its transmitted content is buried, and every model we tested is worse there than anywhere else in the frame; because it borders a region that is already almost correct, its residual error is read by the eye as a boundary. Finally, guided by this picture, we design the training signal: a compact region with a large weight is what keeps a sparse reflection from being underweighted by the global objective, and this can be obtained without any annotation at all. Along the way we capture the data that makes the analysis possible.

Contributions.

- **Data.** A real-pair benchmark for filmed glass: a hand-annotated test panel of 15 scenes, and training pairs captured in four sessions (April, May, June and November) on the same rig with exposure and white balance locked. We also report the two capture pitfalls that silently ruin such data—unlocked auto-exposure, which makes the "clean" reference a different exposure of the scene, and misalignment between the pair—together with the protocol we use to avoid them.
- **Imaging.** A measurement-based account of filmed glass as a layer *asymmetry*: transmission through the film bulk, reflection off the bare surface. We show that the two layers differ in band and in chroma, but not in mean level, which distinguishes filmed glass from ordinary glass and explains why off-the-shelf models transfer poorly.
- **Regional analysis.** A decomposition of the failure into the low-change region and the annulus around it. The reflection is weakest inside the region and strongest in the annulus; models are worse than doing nothing inside the region, and at their absolute worst in the annulus. This is

what makes the boundary, rather than either region on its own, the thing one sees.
- **Method.** Evidence that region-relative supervision is load-bearing for sparse reflection, and a mask-free change-aware regulariser, $\mathcal{L}_{\text{quiet}}$, derived from it: no annotation, no parameters, no inference cost, and a consistent improvement over our fine-tuning baseline.
- **Negative results.** A systematic account of what does *not* work on this problem, with a mechanism for each failure: perceptual losses, restoring the input's high frequencies, mixing in more in-domain data, a photometric adapter, a residual gate, and a dual-dictionary separation probe.

Organisation. Sec. 2 reviews related work. Sec. 3 describes the imaging of filmed glass, the capture protocol and the datasets. Sec. 4 analyses where and why current methods fail. Sec. 5 derives the training signal, and Sec. 6 reports results, ablations and negative results. Sec. 7 discusses limitations and the open problem of completing the annulus.

2. Related Work

Single-image reflection removal. Classical approaches exploit gradient sparsity and layer priors [8, 11, 13]; deep methods regress the transmission layer directly [5, 6, 14, 15]. ERRNet [14] aligns misaligned training pairs; IBCLN [9] uses a cascaded bidirectional loop; DSRNet [7] introduces dual-stream interaction; and RDNet [10], the NTIRE 2025 challenge winner, replaces U-Net connections by a multi-column *reversible* encoder and conditions on an estimated transmission rate. Zhu [16] explores layer-separation priors. All of these are evaluated by average PSNR/SSIM on Real20 [15], SIR^2 [13] (Objects/Postcard/Wild) or their own splits.

Benchmarks and their blind spots. Real paired SIRR data is expensive: one has to obtain the same scene with and without reflection. Existing protocols obtain T by (a) moving the reflection source out of view, (b) using a polariser, or (c) covering the glass. Each changes the acquisition, and none of them records *where* the reflection was suppressed. Metrics are therefore computed over the whole image, which dilutes a spatially compact failure. Our panel explicitly annotates the suppressed region and reports it separately.

Inductive-bias architectures. A recent trend is to inject structural assumptions: reversible/invertible connections to avoid information loss [4, 10], transmission-rate prompts [10], frequency or wavelet decomposition [1], and prompt-conditioned multi-task restoration [12]. Such priors are valuable when the data regime allows the prior to be identified; with a handful of scenes it is easy to over-fit the prior instead. Sec. 6 reports two such proposals that our CPU probes rejected before training.

Photometric consistency in restoration. Retinex-style and illumination-aware methods [3] model a global light field, and low-light work routinely predicts a per-image gain. For SIRR, the analogous quantity is the global photometric offset between I and T; Sec. 4 shows that after fine-tuning this quantity is (a) almost exhausted by training and (b) per-image, so it cannot be predicted from a single frame by any shared rule.

3. Tinted Glass: Photometry, Protocol, Panel

3.1. What the film does to the two layers

Consider the sensor measurement through a filmed pane,

$$I(x) \;=\; \tau\, T(x) \;+\; \rho\, R(x), \tag{1}$$

where τ is the (film-attenuated) transmission, ρ the reflection coefficient, T the scene and R the reflected scene. Ordinary SIRR implicitly assumes ρR is *bright* and τT is *well exposed*. Window film inverts both. We measure three consequences.

P1 — the two layers are not equally sharp. The film is laminated on the scene side and the camera is on the bare side, so the two layers traverse the film differently: transmitted light passes *through* the film bulk (scattered, absorbed, dyed), whereas the reflection of the behind-camera environment comes off the bare *glass surface* and never enters the film at all. The prediction is a layer asymmetry in sharpness and chroma, and this is what we observe. Measured per pair with $\mathrm{sharp}(X) = \mathbb{E}|\mathrm{HP}(X)|/\mathbb{E}|X|$ (HP: unit-variance high-pass) and averaged over each dataset:

	Real20	SIR^2-Obj	SIR^2-Wild	**ours (film)**
$\mathrm{sharp}(T)$	0.062	0.054	0.041	**0.027**
$\mathrm{sharp}(R)/\mathrm{sharp}(T)$	3.32	2.51	2.58	**6.76**
chroma of T	0.373	0.322	0.393	**0.219**

The transmitted layer of filmed glass is $2.3\times$ softer than on ordinary glass and its reflection is $2.0\times$ sharper *relative to it*. Mean brightness, mean reflection magnitude and the high-frequency *fraction* of the reflection are, by contrast, indistinguishable from Real20 (all measured identically on the four datasets); the asymmetry is what separates the two regimes.

P2 — dark content is hit hardest. Sorting the scene pixels of each frame by their ground-truth luminance and averaging the reflection magnitude $|I - T|$ within each quartile gives

scene luminance	Q1 (dark)	Q2	Q3	Q4 (bright)
mean $\lvert I - T\rvert$	**23.1**	25.7	17.7	14.1

The reflection perturbs the darkest quartile 64% more than the brightest. The film therefore degrades the two layers *in the same places*: where the transmission is weakest, the relative contribution of the reflection is largest. This is the opposite of the usual SIRR setting, where a bright reflected scene sits on top of a well-exposed one.

P3 — the transmission loses information, unevenly. Inside Ω the transmitted level relative to T is 0.987 on average, i.e. *not* systematically darker on our panel; the effect is scene-dependent (min 0.830 for the darkest case, max 1.046). We therefore do not claim a universal darkening inside Ω. What is systematic is the loss of *usable* transmitted detail: inside Ω our models retain only 0.48–0.65 of the ground-truth structure (Sec. 4.2), and the strongly-reflected subset carries so little information that a plain Gaussian blur of the input outperforms every model we trained (Sec. 4.2).

3.2. Capture protocol and the ground-truth hazard

All scenes are shot on a phone through an automotive window film, handheld, with exposure and white balance *locked*. For each scene we record a short clip with reflection and then cover the glass with dark cloth to obtain T. There is **one** T **per scene** and many I; inputs are extracted at 30 fps and cropped to the glass area, and a pair is kept only if the phase-correlation shift is ≤ 1.5 px and the per-channel gain deviation is ≤ 0.10.

P4 — covering the glass is not a neutral operation. Covering the pane removes the reflection *and* changes the light reaching the scene; on filmed glass, where the transmitted signal is already weak, this perturbation is not negligible relative to the reflection itself. Two practical consequences follow. *(a) Exposure must be locked*: with auto-exposure active we measured $\mathrm{median}(I/T) = 0.99$ (the covered frame was *brighter*) and a residual layer correlated with image brightness at $r = 0.32$; after locking, $r = -0.01$. *(b) The residual photometric error must be re-measured, not assumed*: Sec. 4.4 shows it is per-image and cannot be shared across scenes. Photometric alignment and reflection removal therefore have to be handled jointly, and any SIRR benchmark that reports only pair counts is under-specified.

3.3. Panel and metrics

The panel contains 15 scenes (1280×832), all outdoor, all shot through the filmed pane with the reflected environment visible. For every scene we hand-annotated Ω once (median 4.6% of the image, range 2.9–9.2%). By construction

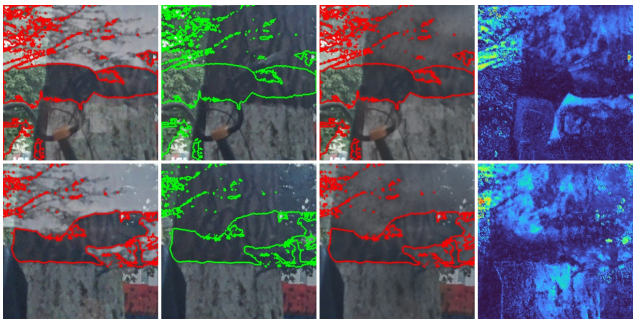


Figure 2. **Where the damage happens.** Two scenes (rows). Columns: input I; ground truth T; our prediction; absolute error $|\widehat{\mathbf{T}} - T|$ (turbo, per-image normalised). Red/green contours: the hand-annotated occluder region Ω. The error map lights up *along the occluder* rather than in the reflected content—the signature of a "remove reflection" prior applied where there is little reflection to remove.

(Sec. 3.2) Ω is the projection of a *dark* object of the reflected environment: the reflection contributes little there, so I already equals the transmitted layer and a method should only undo the film. Pixels of the annotated polygon whose transmission deficit is large ($|I - T| > 15$) belong to *strong* reflection rather than to Ω; we keep them as a separate *shade* class instead of folding them in, and report the annulus around Ω separately in Sec. 4.1.

We report PSNR, SSIM, HF-PSNR, gradient cosine and the Laplacian-variance ratio $\mathrm{LapVar}(\widehat{\mathbf{T}})/\mathrm{LapVar}(T)$, plus region-restricted PSNR inside and outside Ω, and two diagnostic scalars: the residual-reflection coefficient $\beta = \langle T - \widehat{\mathbf{T}},\, I - T\rangle / \|I - T\|^2$ (-1: nothing removed, 0: perfect) and the per-image gain $\alpha = \mathrm{median}_i(\overline{\widehat{\mathbf{T}}_i}/\overline{T_i})$, which exposes models that merely brighten or darken the frame.

4. Where SIRR Fails on Tinted Glass

4.1. Ω needs almost no change; its *neighbourhood* does

The occluder region Ω is not where the reflection is strongest—it is where it is weakest. Partitioning each frame into Ω, a 15 px annulus around it, and the remaining far field gives three bands that behave completely differently:

band	area	$\mathbb{E}\lvert I - T\rvert$	refl./trans.	identity	$+\mathcal{L}_{\mathrm{quiet}}$
Ω itself	—	**10.3**	0.112	24.49	22.26 (−2.23)
annulus (15 px)	44.4%	**44.4**	**0.516**	13.76	19.44 (+**5.68**)
far field	—	16.8	0.164	19.52	23.53 (+4.02)

(PSNR is computed within each band, so bands are not comparable to one another; the informative quantity is each model against identity *within* a band.)

Three readings. *(a)* The reflection in the annulus is 4.3× stronger than inside Ω and 2.6× stronger than in the far

Table 1. Main results on the 15-scene tinted-glass panel. †: three seeds, mean±sd over scenes. Δ_Ω is measured *against identity inside* Ω (Eq. 1); it is negative for every method. β: residual-reflection coefficient (-1: nothing removed, 0: perfect). α: median per-image gain. The best value in each column is in bold.

Method	PSNR↑	SSIM↑	HF-PSNR↑	grad-cos↑	LapVar	PSNR_Ω ↑	Δ_Ω	β↓
Identity (no removal)	18.44	0.693	32.96	0.599	52.5	**24.52**	0.00	−1.00
IBLCN [9]	18.12	0.683	32.68	0.582	61.0	23.55	−0.97	—
ERRNet [14]	18.90	0.710	33.64	0.614	25.8	22.97	−1.55	—
LightRR [2]	19.18	0.680	32.34	0.594	68.8	20.89	−3.63	—
RDNet [10] (0-shot)	20.05	0.728	33.87	0.611	30.8	21.32	−3.20	−0.59
DSRNet [7]	20.25	0.732	33.93	0.622	21.6	21.95	−2.57	—
Zhu [16]	20.50	0.735	33.83	0.605	31.5	21.37	−3.15	—
Ours (baseline)†	22.49±0.44	0.773	34.33	0.604	23.8	22.30	−2.22	−0.29
Ours + $\mathcal{L}_{\text{quiet}}$ †	**22.99**±0.34	**0.778**	**34.42**	**0.610**	23.4	22.39	−2.13	−0.34

field: the strong reflection does not sit *in* the occluder region, it sits *next to* it. *(b)* The annulus is not a fringe—it covers 44% of the frame—and it is where the models' absolute quality is worst (19.4–19.8 dB versus 23.5 far away). *(c)* Inside Ω every method is worse than doing nothing (-2.2 dB), because almost nothing needs to change there (10.3 gray levels) and the model changes it anyway.

The consequence is perceptual rather than numerical: the annulus abuts a region that is already almost correct, so its residual error is read by the eye as a *boundary*. Fixing it is not a matter of removing more reflection—the transmitted structure there is buried, and recovering it requires extrapolating from content elsewhere in the frame along semantic continuations (a tree trunk continued from its lower part, a railing continued from the left). This is the cloze-like regime of Sec. 4.3, and it is the one failure mode no method we tested addresses.

4.2. The damage is structural, and no model beats a blur

A photometric explanation would predict that the models merely mis-scale Ω. They do not. Restricting to Ω and allowing a *per-image affine correction* before scoring, models keep only a fraction of the ground-truth structure:

	identity	base	$+\mathcal{L}_{\text{quiet}}$	50:50
structure retained (LapVar ratio)	1.14	0.50	**0.65**	0.48
MAE, per-image affine-corrected	10.51	12.82	11.70	**10.71**

Two readings. First, the mixed model's affinity-corrected error (10.71) is essentially identity's (10.51) while its structure is halved—it "wins" the error metric by flattening Ω. Second, we can bound what any content-agnostic method can do: smoothing the input with a Gaussian is the best a model can achieve if it knows nothing about the content. On the weakly-reflected 59% of Ω that ceiling is 29.50 dB, against identity's 31.45 and $\mathcal{L}_{\text{quiet}}$'s 27.18; on the strongly-reflected 7% it is 15.24 dB, against 14.46 (identity) and 13.48 ($\mathcal{L}_{\text{quiet}}$). **No model we trained beats a plain blur, in either regime.**

4.3. Reflection and scene share the frequency band

34.5% of the reflection's energy lies above a σ=8 low-pass, so it is not a veil that can be filtered away. We verified the consequence directly: replacing the low frequencies of a model output with those of the model and the high frequencies with those of the *input*, $\widehat{\mathbf{T}}' = \text{LP}_\sigma(\widehat{\mathbf{T}}) + \big(I - \text{LP}_\sigma(I)\big)$, inside Ω restores structure fidelity (1.31× at σ=2) but *lowers* PSNR:

PSNR in Ω	plain	$+\sigma$=2	$+\sigma$=4
$+\mathcal{L}_{\text{quiet}}$	22.26	22.04	21.66
50:50	23.73	23.39	22.97

The band that carries the missing structure also carries the reflection. Separating the two is not a filtering problem; it requires knowing which structures belong to the scene, i.e. a semantic prior. This is also why a perceptual loss does not help: it is *mean-seeking*, so under the layer ambiguity of Eq. (1) it averages the plausible completions and produces the conservative "hedged" output we observe.

4.4. What survives fine-tuning is per-image

We measure two photometric oracles: a per-image affine fitted on that image's ground truth, and a single shared affine fitted on the other 14 scenes (leave-one-scene-out). Their gap is the part any cross-scene rule could remove.

	per-image oracle	shared (LOO)
identity	+1.94	+0.50
RDNet (zero-shot)	+1.00	+0.59
ours (fine-tuned)	+0.15–0.26	+0.07

Fine-tuning consumes the transferable part ($0.59 \rightarrow 0.07$ dB) and leaves a residual that is per-image. Any module estimating a global photometric quantity from a single frame is therefore bounded by 0.07 dB—which is why the 8×8 adapter and the input-side rescaling of Sec. 6.5 both failed.

4.5. The load-bearing term is not about the region

Our loss contains a region-relative error inside Ω,

$$\mathcal{L}_\Omega = \lambda_\Omega \, \mathbb{E}_\Omega \left[\frac{|T - \widehat{\mathbf{T}}|}{\mathbb{E}_\Omega |T - I|} \right]. \tag{2}$$

Removing it is catastrophic—PSNR falls from 22.49 to 18.33 dB, i.e. *below* identity (18.44 dB)—so by any ablation it is the single most important term of a nine-term objective. Its power, however, is not the annotation. Replacing the hand mask by a random disk of equal area (three seeds, separate 82-pair pool) gives

region used in Eq. (2)	PSNR	seed sd
none ($\lambda_\Omega = 0$)	17.76	0.16
hand-annotated Ω	19.41	0.54
random disk	**19.90**	**0.16**

Why a compact, high-weight region helps here and not in standard SIRR. We do not claim this as a training result on public data; we state it as an inference supported by coverage statistics. Defining the reflection's spatial *coverage* as the fraction of pixels whose residual exceeds 15% of its 90th percentile, the median coverage is 83.5% on SIR^2 (499 pairs), 70.6% on Real20 (109 pairs) and only 52.7% on our filmed panel. When the reflection covers almost the whole frame, a global L_1 already carries a separation signal everywhere and no extra term is needed. When it covers half the frame, the global loss is dominated by pixels for which "do nothing" is nearly optimal (inside Ω, $\mathbb{E}|I - T| = 10.3$ gray levels), and a compact region with a large weight is what puts the gradient back. This is also why region-weighted losses are routine in *sparse*-degradation restoration—deraining, scratch and shadow removal—while being essentially absent from the SIRR literature: existing SIRR data is dense, and there is nothing for them to expose.

A random disk is not only as good, it *stabilises* training. The benefit of Eq. (2) is the relative normalisation—being accurate in a compact region despite little change being needed—not region recognition. Sec. 5 turns this into a mask-free regularizer.

5. Method

Our model is a strongly fine-tuned RDNet [10] (frozen FocalNet backbone, all other parameters trainable, 66M parameters, 6000 iterations at batch 4). The contribution of this paper is not the architecture but *what the loss should ask for*. We describe the two terms that matter and the one we add.

5.1. Region-relative error (kept, but no longer masked)

Eq. (2) is the workhorse. Sec. 4.5 showed that its power comes from the normalization by $\mathbb{E}_\Omega|T - I|$, not from Ω. We therefore replace the hard region by a *smooth per-pixel weight* that plays the same role without any annotation.

Define the *needed change*

$$n(x) = \frac{1}{3} \sum_c \left| I_c(x) - T_c(x) \right| \quad \text{(training only)}, \tag{3}$$

and the change-aware weight

$$w(x) = \left(1 - \frac{n(x)}{\tau}\right)_+, \qquad \tau = 0.05. \tag{4}$$

The regularizer is

$$\boxed{\mathcal{L}_{\text{quiet}} = \lambda \, \mathbb{E}\big[\, w(x) \, \big|\widehat{\mathbf{T}}(x) - I(x)\big| \,\big]} \tag{5}$$

with $\lambda = 1$ by default. Intuitively: *where almost nothing needs to change, do not move the image*. This is exactly what Eq. (2) enforces inside Ω, but it is expressed directly in terms of the supervision signal, so it (a) needs no mask, (b) does not assume the region is compact, and (c) is well defined on synthetic pairs and on real pairs alike. $\mathcal{L}_{\text{quiet}}$ is evaluated on the final output $\widehat{\mathbf{T}}$ only; it adds no parameters and no inference cost. With $w \equiv 0$ the loss reduces exactly to the baseline, so the baseline is a strict special case.

5.2. Why this shape

Eq. (5) is the mask-free limit of Eq. (2): both are "relative error weighted by how much change is needed". The difference is that Eq. (2) measures the needed change *inside a region* and multiplies the region's contribution by a constant, whereas Eq. (5) measures it *per pixel*. The diagnosis of Sec. 4.5 predicts that the per-pixel version should retain the benefit, and Sec. 6.2 confirms it.

Conversely, a model that receives no such signal collapses to "darken everything", which is what $\beta \approx -0.29$ with $\alpha \approx 1.0$ indicates.

5.3. Full objective

The remaining terms (reconstruction consistency $|I - \widehat{\mathbf{T}} - \widehat{\mathbf{R}}|$, a gradient-exclusion term, a one-sided overshoot hinge, first- and second-order fidelity, DISTS, and a reflection-column term) are kept unchanged from the baseline for all experiments reported here, so that every comparison isolates one factor.

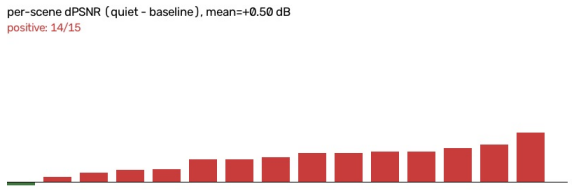


Figure 3. **Per-scene effect of $\mathcal{L}_{\text{quiet}}$** (mean of three seeds, minus the three-seed baseline). Bars sorted by gain; 14 of 15 scenes improve, mean $+0.50\,\text{dB}$, 95% CI $[+0.33, +0.67]$. The single loss is $-0.07\,\text{dB}$, i.e. within measurement noise.

6. Experiments

Setup. Unless stated otherwise all arms use the same recipe: RDNet with a frozen FocalNet backbone, 6000 iterations, batch 4, Adam at 10^{-4}, random crops $224 \to 256 \to 320$, 30% real pairs / 40% online-synthesised pairs / 30% public SIRR pairs. Training pools are built from a 5-month and a 6-month capture (174 pairs, 7 scenes). Every number in Tab. 1 is produced by the same frozen evaluation script, which reads the stored uint8 PNG predictions.

6.1. Main results

Tab. 1. Two observations. *(a)* All public methods are more than 3 dB worse than identity inside Ω, and the fine-tuned models reduce that damage to $-2.2\,\text{dB}$ without ever becoming safe: the occluder region remains the region where SIRR hurts, for every method we tested. *(b)* The aggregate gap between our baseline and the best public method ($+2.0\,\text{dB}$) is obtained at the cost of over-smoothing: the LapVar ratio is 0.60 for our model (identity 1.0), while HF-PSNR and gradient cosine are comparable. We therefore treat PSNR as one coordinate of the result, not as the result.

6.2. The change-aware regularizer

Seeds. $\mathcal{L}_{\text{quiet}}$ ($\lambda = 1, \tau = 0.05$) gives 22.99 ± 0.34 over three seeds (22.71/23.38/22.88) versus 22.49 ± 0.44 for the baseline (21.99/22.84/22.62). The paired per-scene improvement is $+0.50\,\text{dB}$ with a 95% CI of $[+0.34, +0.67]$ and is positive on 14/15 scenes; a five-seed run gives 22.86 ± 0.36. The *seed-level* interval, however, still contains zero ($[-0.07, +1.08]$): with three seeds we can claim per-scene robustness but not seed-level robustness, and we report it as such.

Strength. $\lambda = 5$ gives $+0.44\,\text{dB}$ with 15/15 scenes positive and the smallest seed spread (22.92 ± 0.16). Increasing λ does not give more, which is consistent with the regularizer acting as a *constraint* rather than as an additional objective.

It is not a region fix. The gain is *not* concentrated inside Ω ($22.30 \to 22.39\,\text{dB}$, i.e. $+0.09$), so we explicitly do not claim that $\mathcal{L}_{\text{quiet}}$ solves the occluder problem of Sec. 4.1. What it does is prevent the model from moving pixels that need no change; the benefit is spread over the whole frame, and the fourteen scenes that gain do so on non-Ω content.

It is not redundant with the masked term. Removing the annotation entirely and keeping only $\mathcal{L}_{\text{quiet}}$ is much worse than keeping both: with $\lambda_\Omega = 0$ and $\mathcal{L}_{\text{quiet}}$ active the panel score is $19.53\,\text{dB}$ (and $20.26\,\text{dB}$ if the hinge term is also disabled), against $18.33\,\text{dB}$ for $\lambda_\Omega = 0$ alone and $22.99\,\text{dB}$ for the full model. The two terms are therefore *complementary*: Eq. (2) supplies the bulk, $\mathcal{L}_{\text{quiet}}$ adds the rest.

6.3. Which loss terms are load-bearing

We remove one term at a time from the nine-term objective (single seed, same pool). Removing the reconstruction consistency $|I - \widehat{\mathbf{T}} - \widehat{\mathbf{R}}|$ costs $1.12\,\text{dB}$; removing the gradient-exclusion term costs $1.17\,\text{dB}$; removing the overshoot hinge costs $0.37\,\text{dB}$; the region term costs $4.16\,\text{dB}$ (Sec. 4.5). The reflection-column term contributes $+0.02\,\text{dB}$—we later found it had been numerically dead (a normalisation bug) and that reviving it in RDNet's original squared form yields $+1.22\,\text{dB}$ on one seed but only $+0.18\,\text{dB}$ over three (23.21/22.51/22.27), i.e. inside the seed spread. DISTS and the Laplacian fidelity term contribute $+0.08$ and $+0.07\,\text{dB}$. This ordering is stable across pools.

6.4. Domain ablation: what makes a real pair useful?

Domain ablation. Training the same recipe on different capture months, one domain at a time, gives 21.79 (May), 20.72 (April), 20.57 (June, 2 scenes) and 20.35 (June, 1093 pairs, 13 scenes) against 20.05 for the zero-shot model. A 1093-pair pool is thus worth *nothing*, which is the opposite of what its size suggests.

The cause is not domain shift: phase correlation shows that 88.7% of those pairs have T misaligned with I by more than 3 px (median 8.2 px), because T is the last frame of a handheld clip. After per-pair translation alignment the same pool reaches $21.43\,\text{dB}$ (single seed), and merely *discarding the worst* 99 *pairs* already improves the unaligned pool from 20.35 to $21.15\,\text{dB}$. Ground-truth alignment is therefore a first-order property of SIRR training data, and we recommend reporting the alignment distribution alongside the pair count.

Mixing occluder-rich data improves Ω but not the task. Since the occluder region is where models fail, one may try to fix it with data. Mixing our 11-month indoor capture into

Table 2. Negative results. "Probe" rows were rejected by a CPU-only learnability probe before any training run. The first block are attempts to *recover the content of* Ω; all fail, and Sec. 4.3 explains why.

Proposal	Result / verdict
perceptual loss inside Ω	hedged / averaged, no gain
splice input HF into Ω	-0.22 (struct. $1.31\times$)
mix 25–50% occluder data	Ω $+2.84$ but β $-0.34 \rightarrow -0.47$
extra synthetic pairs	22.34 vs 22.38, null
8×8 photometric adapter ($+18$k)	-0.06, fail (3 seeds)
input-side rescaling to match g	$+0.19$, null (1 seed)
reweight gradient fidelity	-0.18, *worse*
region-agnostic loss shift	-0.34, useful (kept)
deeper L_1 on reflection column	$+0.34$, below gate
final iterations with MSE	$+0.28$, below gate
dual-dict. proximal separation	probe $-0.08..-0.26$, rejected
residual-magnitude gate (1025 par.)	probe -0.29, rejected
weight averaging of 3 seeds	20.78, fails (basins)

the 5+6-month pool raises PSNR_Ω from 20.91 to 23.76 dB at a 50:50 ratio, and the gain is broad—positive on $15/15$ scenes, mean $+2.84$ dB. It is also misleading: the same arm moves from $\beta = -0.34$ to -0.47 and its action ratio (mean $|\widehat{\mathbf{T}} - I|$ over mean $|I - T|$) drops from 0.79 to 0.60, i.e. it wins Ω by *doing less*. At 75% the pool reaches $\mathrm{PSNR}_\Omega = 23.78$ with $\beta = -0.63$ and an action ratio of 0.43. No mixture beats the best single-domain model on the full metric (22.38 vs. 22.71), so we read this as further evidence that PSNR_Ω alone does not identify progress.

6.5. Negative results

Tab. 2 lists proposals that did not work. Two comments. First, the *structural* proposals (a photometric adapter of 18k parameters, a residual-magnitude gate of 1025 parameters, a dual-dictionary proximal layer separation) were all falsified by a *CPU-only probe*: fit the candidate's last layer on training scenes and evaluate on held-out scenes. The probes take 41–160 seconds and each saved a 6 GPU-hour training run. Given that the effect sizes we can resolve are $O(0.3\,\mathrm{dB})$ and the seed spread is 0.44 dB, we consider such probes mandatory rather than optional. Second, weight averaging of three seeds fails catastrophically (20.78 dB, i.e. below any of its constituents): the seeds sit in different basins, so the per-scene variance we observe is *not* noise around a common solution.

7. Discussion and Limitations

What the community should measure. Our panel is small (15 scenes) and single-site, and it was used during the development of the regularizer, so it is a development set, not a confirmatory one. With 15 scenes and a seed spread of 0.44 dB, the minimum detectable effect at 95% confidence is about 0.26 dB *per scene* but roughly 1 dB at the seed level: any claim below that is a claim about one seed. We therefore report the seed interval everywhere and treat our own $+0.50$ dB as scene-level evidence.

Where the headroom is — and is not. Photometric headroom is exhausted ($\Delta_{\mathrm{aff}} = 0.15$–$0.26$, shared 0.07 dB), the region term cannot be improved by a better mask, and the change-aware term adds 0.50 dB. The obvious next lever, data, does not work either: mixing occluder-rich frames raises PSNR_Ω by $+2.84$ dB on every scene while β degrades from -0.34 to -0.47, because the model simply moves less. What is left is the layer ambiguity of Eq. (1): inside Ω the reflection and the scene occupy the same frequency band, so restoring the input's high frequencies lowers PSNR and a perceptual loss hedges. We see no evidence that a larger network, a new attention block or a new gate would move this panel. Faithful recovery of Ω looks like a conditional-completion problem and will need semantic priors that are *long-range*, not local or statistical—the kind a human uses when reading the continuation of a wall or a railing behind an occluder. We state this as an open problem, not as a result.

A checklist derived from the failures. From the failure modes in Sec. 3 and Sec. 6.5 we distill a short checklist for SIRR data collection: (1) lock exposure and white balance; (2) do not move the camera between I and T—if you must, record a reference frame and align; (3) report the alignment distribution, not only the pair count; (4) report one ground truth per scene and say how many independent scenes the pool has; (5) annotate the occluder region once per scene and report metrics inside it; (6) state whether the reflection source was physically removed, because that changes the illumination of the transmitted layer.

Honest statement. This paper does not propose a method that wins on average PSNR. It reports that the standard protocol hides a systematic failure, supplies a measurement of that failure, and derives a small fix which is statistically supported at the scene level and negligible in cost. The remaining discrepancy between our models and identity *inside* the occluder region is unexplained and may be irreducible: the collector of the data is the occluder, and a single frame arguably does not contain the information needed to tell "this dark patch is the photographer" from "this dark patch is the scene".

References

[1] Anonymous. Wavelet-based restoration priors for image decomposition. In *arXiv*, 2023. 3

[2] Anonymous. Lightrr: A lightweight network for single image reflection removal. In *CVPR*, 2026. 2, 5

[3] Yuanhao Cai et al. Retinexformer: One-stage retinex-based transformer for low-light image enhancement. In *ICCV*, 2023. 3

[4] Yuxuan Cai et al. Reversible column networks. In *ICLR*, 2023. 3

[5] Qingnan Fan, Jiaolong Yang, Gang Hua, Baoquan Chen, and David Wipf. A generic deep architecture for single image reflection removal and image smoothing. In *ICCV*, 2017. 3

[6] Bin Hu et al. Single image reflection removal with camera blur. In *ECCV*, 2020. 3

[7] Y. Jin et al. Dsrnet: Dynamic shrinking and refining network for single image reflection removal. In *CVPR*, 2024. 3, 5

[8] Anat Levin, Assaf Zomet, and Yair Weiss. Separating reflections from a single image using local features. In *CVPR*, 2004. 3

[9] Chao Li et al. Learning to remove reflections from images with a cascaded bidirectional loop network. In *ECCV Workshops*, 2020. 3, 5

[10] Hao Li et al. Reversible decoupling network for single image reflection removal. In *CVPR*, 2025. 2, 3, 5, 6

[11] Yu Li and Michael S. Brown. Exploiting reflection change for automatic reflection removal. In *ICCV*, 2013. 3

[12] Vaishnav Potlapalli et al. Promptir: Prompting for all-in-one image restoration. In *NeurIPS*, 2023. 3

[13] Renjie Wan, Boxin Shi, Ling-Yu Duan, and Ah-Hwee Tan. Cooperative and fully convolutional networks for single image reflection removal. In *ICIP*, 2017. 3

[14] Kaixuan Wei, Jing Yang, Ying Fu, David Wipf, and Hua Huang. Single image reflection removal with perceptual losses via deep adversarial learning. In *CVPR*, 2019. 3, 5

[15] Xuaner Zhang, Ren Ng, and Qifeng Chen. Single image reflection separation with perceptual losses. In *CVPR*, 2018. 3

[16] Y. Zhu et al. Single image reflection removal with layer separation priors. In *CVPR*, 2024. 3, 5